MOHAMMAD RAOUF ABEDINI

AI Security Researcher · Adversarial Robustness & AI Control · Empirical Safety Evaluation

Sydney · Relocating to San Francisco, CA · Visa sponsorship required

raoof.r12@gmail.com linkedin.com/in/mohammad-raouf-abedini-885a9226a

github.com/Raoof128 · raoufabedini.dev · ORCID 0009-0000-6214-258X

PROFESSIONAL SUMMARY

Security researcher focused on keeping tool-using AI agents safe and beneficial: builds host-side controls for what an agent is permitted to do and stress-tests them with matched-control experiments against live models. Sole author of Project Simurgh, an open-source agent-authorization prototype whose October 2026 study reports a benefit on one model and a null result on another. Evaluated Claude for exploitable code and guardrail circumvention in Anthropic's safety-evaluation program via Alignerr. Five sole-author preprints and two coordinated vendor disclosures (Microsoft, Apple) after seven years in security operations and engineering. Available full-time from December 2026 for a three- to six-month San Francisco placement.

RESEARCH & TECHNICAL SKILLS

Empirical AI safety and alignment: matched-control experiment design · adversarial robustness and AI control for tool-using agents · prompt-injection and authority containment · guardrail and classifier evaluation · uncertainty reporting · host-held human-oversight state (delegation, stop, lease)

Large language models (LLMs) and agent tooling: Claude API and Claude Code · OpenAI Responses API with tool calling · self-hosted Llama (3.2-3B, 3.3-70B) · Llama Guard 4 · AgentDojo · MCP servers

AI security, safeguards and red-teaming: guardrail-circumvention analysis · red-team scenario design · vulnerability research and coordinated disclosure · offensive proofs of concept · intrusion detection · phishing-detection ML · incident response · MITRE ATT&CK

Languages and systems: Python · TypeScript · C · C++17/20 · Rust · Kotlin · Lean 4 · SQL · Bash · Docker · GitHub Actions · PostgreSQL · Linux · eBPF

RESEARCH & OPEN SOURCE

Project Simurgh — Inspectable Authorization and Containment for AI Agents (Sole Author) May 2026 – Present

github.com/Raoof128/Project-Simurgh · 92 releases · software DOI 10.5281/zenodo.21737792

- **Designed and ran a transfer study (Oct 2026):** 24 tasks, two trials each, three matched controls (credential-only, composed, deny-all) and two models; 288 replayed branches. On Meta Llama-3.2-3B, composed host controls cut restricted-state trials with new effects from 17/24 to 0/24, kept 23/24 useful completions and blocked no valid proposal. On OpenAI gpt-6-luna the baseline was already 0/24, so the study reports no incremental protection there. Protocol to published package took three days (15–17 Oct 2026).
- **Reported uncertainty per cluster:** the 24 restricted trials fall into six states, so intervals are per state (Wilson 95%: 43.6–97.0% versus 0.0–39.0%); part of the gap holds by construction.
- **Detected and mitigated a defect in the live OpenAI integration:** an eight-case Responses API workflow exposed a role-placement error (8/8 declined). Retained the original traces; a frozen developer-role correction completed 2/3 authorized workflows and declined 5/5 restricted ones, with one refusal still open.
- **Captured a live Llama Guard 4 12B input classifier** over a 180-case set (frozen stage, 2026) and produced containment evidence for all 138 malicious cases it missed, 120 of them downstream injections outside an input-only classifier's view; a coverage claim, not a ranking of classifiers.
- **Scoped AgentDojo results:** live Llama-3.3-70B ASR 9/140 to 0/140; benign utility 7/10 to 6/10. A separate deterministic four-suite run (no real LLM) recorded ASR 0/949, utility under attack 949/949 and benign utility 97/97. These frozen 2026 experiments do not establish general agent safety.
- **Reproducible evidence:** current study archives use SHA-256 manifests; earlier attestations use Ed25519. Replay needs native Linux x86_64/Python 3.12 and installed dependencies. 39 Lean 4 model files have a no-sorry gate; they do not prove implementation correctness. 5,171 tests passed in the 19 Oct 2026 smoke run.
- **Published a self-audit:** 919 project documents reviewed; all 141 dated findings, including earlier overstatements, are in a public finding register linked from the README.

The Invisible Window — Screen-Capture Evasion in WebRTC Proctoring (Sole Author) Feb 2026 – May 2026

github.com/Raoof128/invisible-window-research · IEEE-format preprint v2.0 · DOI 10.5281/zenodo.20376495

- **Screen-capture findings:** 100% evasion in tested Windows 10/11 and macOS 14-26 laboratory configurations. Simulated proctoring, not commercial products; Linux untested. macOS 26 tests used native capture tools rather than browser getDisplayMedia().
- **Coordinated disclosure:** Disclosed to MSRC (Feb 2026) and Apple (Mar 2026); both later classified the behavior as by-design/documented, not a security issue. Windows/macOS PoC source is now public.

Aion-BibleQA — Retrieval and Citation Faithfulness in a RAG System (Sole Author) Jun 2026

Preprint · DOI 10.5281/zenodo.20522874

- **40-question pilot:** 34 answerable questions and 6 refusal cases; Recall@5 = 0.941, mean citation_support = 0.978, 6/6 refusals. One automated judge scored no answer below citation_support < 0.5; two flagged rows scored 0.75 and 0.50. This is not a hallucination-free guarantee.

Further Preprints and Shipped Security Tooling 2024 – 2026

- **Banking Shield** (DOI 10.5281/zenodo.20675513): machine-checked proofs that privacy-sensitive values never reach AI-generated explanations; five-tester dry run, zero leaks.
- **Simurgh device-integrity proofs** (DOI 10.5281/zenodo.20374849): preprint on privacy-preserving device integrity proofs for high-stakes sessions.
- **Student-society event pilot** (DOI 10.5281/zenodo.20549736): 31 consented sessions; ballot choices excluded from records; HMAC-chained audit logs.
- **Mehr Guard:** Kotlin Multiplatform offline URL analysis; 25+ heuristics and 19 red-team cases. Repository-reported 87% F1 on curated URLs and sub-5ms desktop-JVM engine median exclude camera/UI and unmeasured platforms.
- **Systems work:** SentinelSSH, Rust MCP with 17 typed read-only Linux inspection tools and no shell; SentinelFlow, C++17 IDS reporting 28.4M packets/s in a synthetic parser-only benchmark, not full IDS throughput; Zurvan, public Python/Rust MCP knowledge engine. Its 20 Oct 2026 receipt: 3,334 passed, 1 failed, 8 skipped; Z1b closure open.

PROFESSIONAL EXPERIENCE

AI Safety Evaluator (Claude) — Alignerr Jan 2026 – Mar 2026
Anthropic safety-evaluation program · Remote

- Stress-tested Claude's safety guardrails: assessed outputs for exploitable code and harmful patterns, analyzed circumvention methods, and delivered rubric-based findings.

Cofounder — Syllabus-Sync May 2026 – Present
Macquarie University Incubator startup · Sydney · syllabus-sync.app

- **Led security/backend work:** WebAuthn passkeys, TOTP MFA, Postgres RLS and audit logging in a shared Next.js/TypeScript student platform. The 17 Aug 2026 CI test job passed 1,343 tests in 155 files; the separate security job failed, so an overall CI pass is not claimed.

Security Engineer & Full-Stack Developer — Self-Employed (Freelance) Jan 2024 – Present
Sydney

- Delivered secure web applications for commercial clients with OWASP Top 10 controls and OAuth 2.0; 5.0-star rating across 31 Airtasker reviews.

IT Manager & Security Operations — Iran Pharmacy Aug 2019 – Jan 2024
50+ staff · Iran

- Investigated unauthorized-access incidents, built automated detection workflows that cut manual triage by about 30%, and enforced least-privilege access.

EDUCATION

Bachelor of Cyber Security — Macquarie University, Sydney Expected Jun 2027
Final session Feb–Jun 2027 is fully online · Coursework: Natural Language Processing & Machine Learning

Diploma of Information Technology — Macquarie University 2023 – 2024

CERTIFICATIONS & CREDENTIALS

Anthropic (Jul 2026), 20 courses including: Building with the Claude API · Claude Code in Action · Model Context Protocol

Macquarie University micro-credentials (2026, 93–100%): Security of AI · Application of AI · Data Security & Information Privacy · DevSecOps · Identity & Access Management